

Problem_Set3

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3 Problem to demonstrate the role of qualitative (ordinal) predictors in addition to quantitative predictors in multiple linear regression

Consider “diamonds” data set in R. It is in the ggplot2 package. Make a list of all the ordinal categorical variables. Identify the response.

- Run a linear regression of the response on the quality of cut. Write the fitted regression model.
- Test whether the expected price of diamond with premium cut is significantly different from that of the ideal cut.
- What is the expected price of a diamond of ideal cut?
- Modify the regression model in (a) by incorporating the predictor “table”. Write the fitted regression model.
- Test for the significance of “table” in predicting the price of diamond.
- Find the average estimated price of a diamond with an average table value and which is of fair cut.

```
set.seed(1234)
library(ggplot2)

## Warning: package 'ggplot2' was built under R version 4.5.2

library(MASS)

## Warning: package 'MASS' was built under R version 4.5.2

head(diamonds)

## # A tibble: 6 × 10
##   carat cut      color clarity depth table price     x     y     z
##   <dbl> <ord>    <ord> <ord>    <dbl> <dbl> <int> <dbl> <dbl> <dbl>
## 1  0.23 Ideal      E     SI2     61.5   55   326   3.95   3.98   2.43
## 2  0.21 Premium    E     SI1     59.8   61   326   3.89   3.84   2.31
## 3  0.23 Good       E     VS1     56.9   65   327   4.05   4.07   2.31
## 4  0.29 Premium    I     VS2     62.4   58   334   4.2    4.23   2.63
## 5  0.31 Good       J     SI2     63.3   58   335   4.34   4.35   2.75
## 6  0.24 Very Good  J     VVS2     62.8   57   336   3.94   3.96   2.48
```

```

contrasts(diamonds$cut)=contr.sdif(5)
ord_mod=lm(price~cut,data=diamonds)
summary(ord_mod)

##
## Call:
## lm(formula = price ~ cut, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4258  -2741  -1494   1360  15348
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4062.24     25.40  159.923 < 2e-16 ***
## cut2-1        -429.89    113.85   -3.776  0.00016 ***
## cut3-2         52.90     67.10    0.788  0.43055
## cut4-3         602.50     49.39   12.198 < 2e-16 ***
## cut5-4       -1126.72     43.22  -26.067 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3964 on 53935 degrees of freedom
## Multiple R-squared:  0.01286,    Adjusted R-squared:  0.01279
## F-statistic: 175.7 on 4 and 53935 DF,  p-value: < 2.2e-16

summary(lm(price~cut,data=diamonds))

##
## Call:
## lm(formula = price ~ cut, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4258  -2741  -1494   1360  15348
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## Coefficients:
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## (Intercept)   4062.24     25.40  159.923 < 2e-16 ***
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## F-statistic: 175.7 on 4 and 53935 DF,  p-value: < 2.2e-16

summary(lm(price~table,data=diamonds))

```

```
##
## Call:
## lm(formula = price ~ table, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6522  -2751  -1490   1368  15746
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -9109.047    438.450  -20.78  <2e-16 ***
## table        226.984      7.625   29.77  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3957 on 53938 degrees of freedom
## Multiple R-squared:  0.01616,    Adjusted R-squared:  0.01614
## F-statistic: 886.1 on 1 and 53938 DF,  p-value: < 2.2e-16
```

```
aggregate(price~cut,data=diamonds,mean)
```

```
##      cut      price
## 1     Fair 4358.758
## 2      Good 3928.864
## 3 Very Good 3981.760
## 4    Premium 4584.258
## 5     Ideal 3457.542
```

```
head(diamonds$cut)
```

```
## [1] Ideal      Premium    Good        Premium    Good        Very Good
## attr(,"contrasts")
##           2-1  3-2  4-3  5-4
## Fair      -0.8 -0.6 -0.4 -0.2
## Good       0.2 -0.6 -0.4 -0.2
## Very Good  0.2  0.4 -0.4 -0.2
## Premium    0.2  0.4  0.6 -0.2
## Ideal      0.2  0.4  0.6  0.8
## Levels: Fair < Good < Very Good < Premium < Ideal
```

```
lvl=as.numeric(diamonds$cut)
```

```
code_fun=function(x){
  if(x-5==0){
    return(c(0,0,0,0))
  }else if(x-5==1){
    return(c(1,0,0,0))
  }else if(x-5==2){
    return(c(1,1,0,0))
  }else if(x-5==3){
    return(c(1,1,1,0))
  }else{

```

```

    return(c(1,1,1,1))
  }
}
coded_pred=t(sapply(lvl,code_fun))
colnames(coded_pred)=c("Premium","Very_Good","Good","Fair")
diamonds=cbind(diamonds,coded_pred)
ord_mod=lm(price~Premium+Very_Good+Good+Fair,data=diamonds)
summary(ord_mod)

##
## Call:
## lm(formula = price ~ Premium + Very_Good + Good + Fair, data = diamonds)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -4258  -2741  -1494   1360  15348
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   3457.54     27.00 128.051 < 2e-16 ***
## Premium       1126.72     43.22  26.067 < 2e-16 ***
## Very_Good     -602.50     49.39 -12.198 < 2e-16 ***
## Good          -52.90     67.10  -0.788  0.43055
## Fair           429.89    113.85   3.776  0.00016 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3964 on 53935 degrees of freedom
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```